Regularized Multivariate Two-way Functional Principal Component Analysis

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Outline

Introduction

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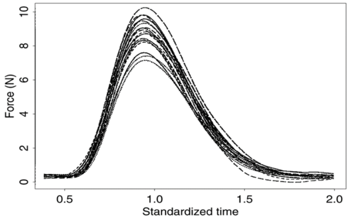

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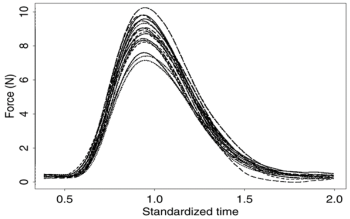

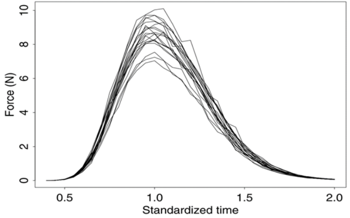

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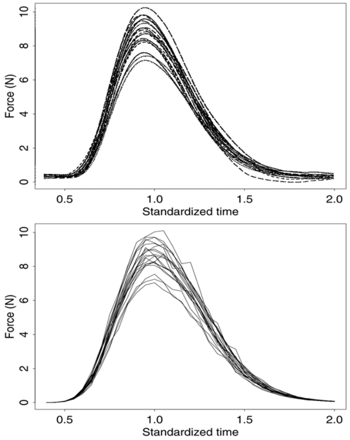
- In practice, these data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- **Functional Data Analysis (FDA)** is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.

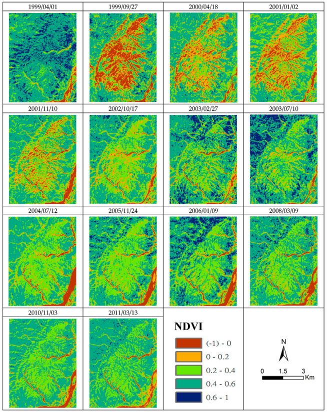


Functional Principal Component Analysis

- **Functional PCA (FPCA):** An extension of classical PCA for dimension reduction and uncovering hidden patterns in functional data; it identifies orthogonal functions that capture the main sources of variation, preserving the most important information. [Ramsay and Silverman, 2005].
- **Extensions of FPCA:**
  - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].
  - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
  - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions [Silverman, 1996, Happ and Greven, 2018].
  - Regularized MFPCA: Penalties improve estimation & interpretability [Haghbin et al., 2023].
- **Impact:** More adaptable, robust, and applicable across diverse scientific and business problems.

Two- way Functional Principal Component Analysis

- **Two-way functional data:** Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.



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- **Extension of FPCA:** Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- **Practical challenges:**
  - Data observed on discrete grids (minutes, hours, days).
  - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- **Proposed framework:**
  - Unified FPCA for two-way multivariate functional data.
  - Smoothness penalties preserve functional structure.
  - Sparsity penalties enhance interpretability.
  - Effective for dimension reduction in complex datasets.

Foundations of FPCA through Minimizing Reconstruction Error

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.
- **Reconstruction problem:**

$$\min_{u,v} \|X - uv^\top\|_F^2 = \mathrm{tr}\{(X - uv^\top)(X - uv^\top)^\top\},$$

- **Optimization steps:**

$$\text{Fix } v: \ u = \frac{Xv}{v^\top v} \qquad \text{and} \qquad \text{Fix } u: \ v = \frac{X^\top u}{u^\top u}$$

Extensions of FPCA via Regularization

- **Goal:** Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^\top\|_F^2 + \mathcal{P}(u, v)$$

- Two directions:
  - **Smooth FPCA:** adds roughness penalty on functions.
  - **Sparse FPCA:** adds sparsity penalty on loadings.
- Algorithms: Based on iterative **power method** and **thresholding** updates.

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- A power algorithm is defined to compute the PCs while incorporating smoothness penalty.
- Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = udv^\top$, v is the first principal component and $u = ud$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

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Algorithm: Penalized Power Iteration

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- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m\times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tunning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\mathrm{CV}(\alpha) = \frac{1}{m}\sum_{j=1}^{m} \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj}\right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j-th diagonal element.

$$\mathrm{GCV}(\alpha) = \frac{1}{m}\frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m}\operatorname{tr}\{S(\alpha)\}\right)^2}.$$

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- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^\top\|_F^2 + p_\gamma(v)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Lasso (Soft-thresholding):**

$$p_\gamma^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow[\text{minimizer}]{} h_\gamma^{\text{soft}}(y) = \text{sign}(y)(|y| - \gamma)_+$$

- **Hard thresholding:**

$$p_\gamma^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow[\text{minimizer}]{} h_\gamma^{\text{hard}}(y) = I(|y| > \gamma)\, y$$

- **SCAD penalty:**

$$p_\gamma^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \dfrac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \dfrac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow[\text{minimizer}]{} h_\gamma^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y| - \gamma)_+, & |y| \leq 2\gamma, \\ \dfrac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

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   3. **Normalize Right Singular Vector:** $v \leftarrow \dfrac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter**: Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

1. Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \ldots, X^K$.

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    2. Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

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3. Select the degree of sparsity as $j_0 = \arg\min\{CV(j)\}$.

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- Trade-off: **Variance explained vs Interpretability vs Smoothness**.

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  - MFPCA $\rightarrow$ **joint modes of variation** across functions
- **Challenges**
  - Discretization & irregular grids $\rightarrow$ noise, missing data
  - High dimensionality $\rightarrow$ unstable eigenfunctions
  - Cross-function correlation $\rightarrow$ requires joint smoothness
- **Proposed Solution: Penalized SVD**
  - **Smoothness penalties**: roughness on derivatives
  - **Sparsity penalties**: ℓ_1/SCAD shrinkage
  - **Block-diagonal roughness matrix** for cross-function structure
- **Impact**
  - Produces **smooth, sparse, interpretable** joint modes
  - More stable & applicable to high-dimensional multivariate FDA

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- **Block structure example**:

$$X_i = \begin{bmatrix} x_{1i}(t_{i1}) & \cdots & x_{1i}(t_{i,m_i}) \\ \vdots & \ddots & \vdots \\ x_{ni}(t_{i1}) & \cdots & x_{ni}(t_{i,m_i}) \end{bmatrix}, \quad \mathbf{X} = [X_1 \quad X_2 \quad \cdots \quad X_p]$$

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- The penalized reconstruction error is

$$\min_{u,v} \|\boldsymbol{X} - uv^\top\|_F^2 + \boldsymbol{\alpha}^\top (v^\top \boldsymbol{\Omega} v),$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^\top$ controls smoothness.

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- The smoothing operator is $\boldsymbol{S}(\boldsymbol{\alpha}) = (I + \boldsymbol{\alpha\Omega})^{-1} \in \mathbb{R}^{M\times M}$.
- The smoothing parameter $\boldsymbol{\alpha}$ is selected via generalized cross-validation (GCV), defined as

$$\mathrm{GCV}(\boldsymbol{\alpha}) = \frac{1}{M}\frac{\|(I - \boldsymbol{S}(\boldsymbol{\alpha}))(\mathbf{X}^T u)\|^2}{\left(1 - \frac{1}{M}\mathrm{tr}\{\mathbf{S}(\boldsymbol{\alpha})\}\right)^2}. \tag{1}$$

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- Algorithm CV Tuning for Sparsity and equation (1) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

- **Data-generating process:** Two functional variables:

$$X_{ij}^{(1)} = u_{i1} v_{11}(t_j) + u_{i2} v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1} v_{21}(t_j) + u_{i2} v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

  - where $u_{i1} \sim N(0, \sigma_1^2)$, $u_{i2} \sim N(0, \sigma_2^2)$, $\epsilon_{ij}^{(k)} \sim N(0, \sigma^2)$, and $n = m = 101$, $t_j \in [-1, 1]$
- **True functional PCs:**
  - Variable 1: $v_{11}(t) = \frac{t+\sin(\pi t)}{s_1}$, $\quad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$
  - Variable 2:

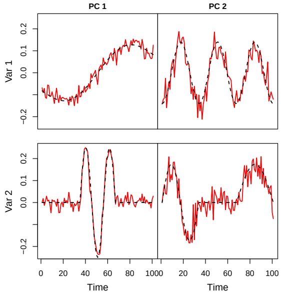
$$v_{21}(t) = \begin{cases} \dfrac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise}, \end{cases} \qquad v_{22}(t) = \begin{cases} \dfrac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \dfrac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise}. \end{cases}$$

Here, s_1, s_2, s_3, s_4 are normalizing constants ensuring unit L^2 norm.

Simulation: Estimation Performance

Scenarios tested:

1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
3. Sparse Multivariate SVD (sparsity penalty)
4. Sparse + Smoothed Multivariate SVD (combined regularization)



(a) MFPCA

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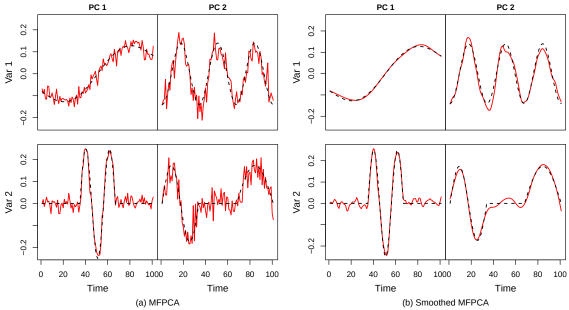

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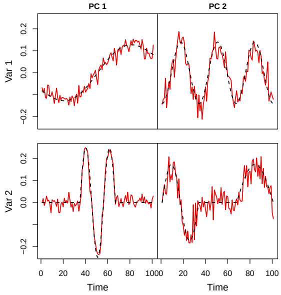

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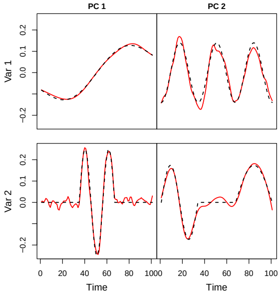
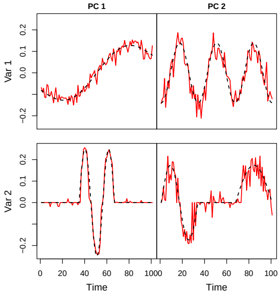


(b) Smoothed MFPCA



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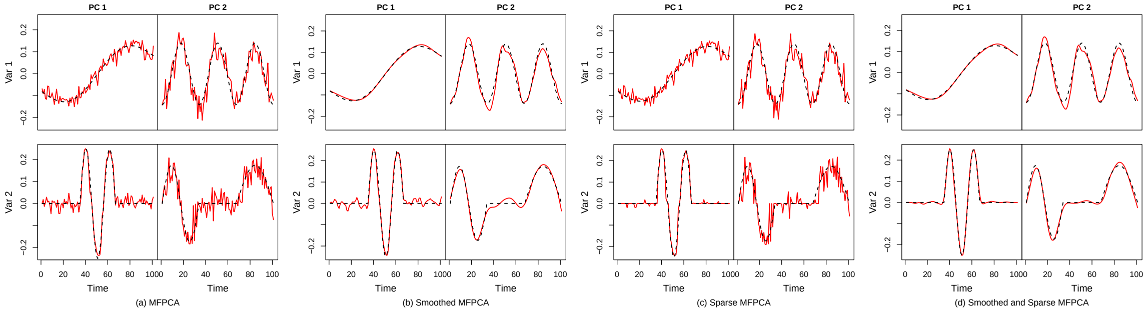


Simulation: Estimation Performance

Accuracy measures:

1. Variable-wise MSE:

$$\mathrm{MSE}_{k\ell} = \frac{1}{m}\sum_{j=1}^{m}(\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

2. Replication-averaged MSE:

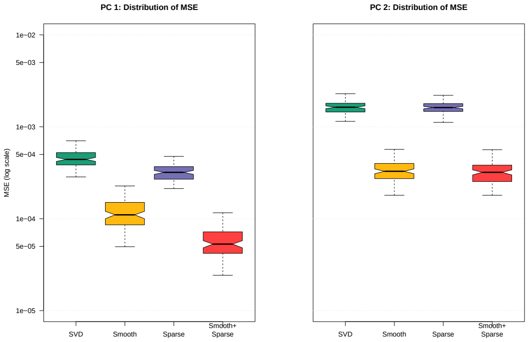
$$\overline{\mathrm{MSE}}_{k\ell} = \frac{1}{R}\sum_{r=1}^{R}\mathrm{MSE}_{k\ell}^{(r)}$$

3. Multivariate MSE:

$$\mathrm{MSE}_{\ell}^{(\mathrm{multi})} = \frac{1}{m}\sum_{j=1}^{m}\sum_{k=1}^{2}(\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

Simulation: Estimation Performance

- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
  - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
  - **Smooth+Sparse** yields lowest error and most stable estimates.
  - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
  - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log10(MSE)

Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	**-4.38**	**-4.28**	**-4.22**	**-4.14**

PC2: Quartiles and Mean log10(MSE)

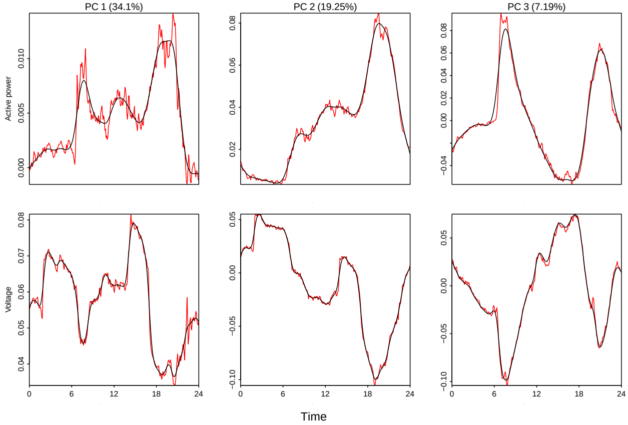
Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	**-3.59**	**-3.50**	**-3.49**	**-3.42**

Application: Household Power Consumption

- **Dataset:** Bivariate functional data including active power and voltage consumption [Hebrail and Berard, 2012] for one household between December 2006 and November 2010.
- **Scaling:** To equalize the contribution of each variable in the multivariate analysis, we rescale them following [Happ and Greven, 2018].

$$\tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i),$$

$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^{m} \widehat{\mathrm{Var}}(X_j(t_i)) \right)^{-1}.$$



First 3 PCs: MFPCA (red) vs ReMFPCA (black)

Regularization reduces noise while preserving the dominant daily consumption patterns, enhancing interpretability without losing key structure.

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  - Results may be rough or overly dense along the unpenalized axis.

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- Multivariate v: $\boldsymbol{\Omega}_v = \mathrm{diag}(\Omega_1, \ldots, \Omega_p).$

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- **Optimization:** Alternate updates of u and v using GCV until convergence → two-way regularized components.

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- **Result:** Robust, interpretable, noise-resistant components for high-dimensional data.

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- Data matrix $\boldsymbol{X}$: seek u, v solving:

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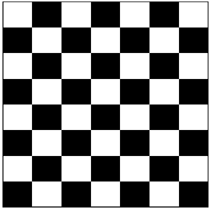
   3. $v \leftarrow v/\|v\|$
3. $\boldsymbol{X} \leftarrow \boldsymbol{X} - \sigma\, u\, v^\top$ to extract the next component.

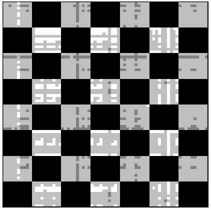
Chess Board

True Chessboard

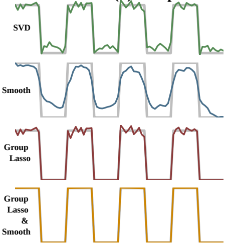


PCA with SVD

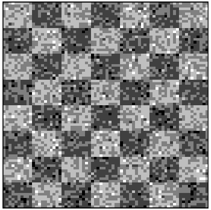
PCA with Group Lasso Penalty



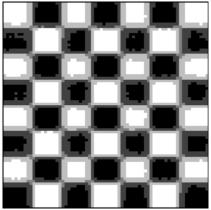
First PC Scores (u) Comparison



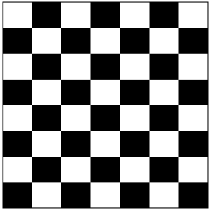
Noisy Chessboard



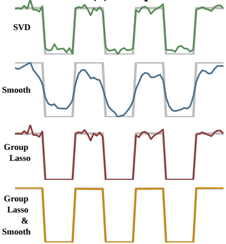
PCA with Smoothness Penalty



PCA with Group Lasso and Smoothness Penalties



First PC (v) Comparison



Selection of Regularization Parameters

- Four sets of tuning parameters:
  - α_u: smoothness of u, γ_u: sparsity of u
  - $\boldsymbol{\alpha}_v$: smoothness of v, $\boldsymbol{\gamma}_v$: sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.
- **Strategy: Conditional tuning**
  1. Initialize all penalties at 0.
  2. Tune γ_u via K-fold CV.
  3. Sequentially tune $\gamma_{v,i}$ using Algorithm~**??**.
  4. With sparsity fixed, tune α_u by GCV.
  5. Tune $\alpha_{v,i}$ using two-way GCV.
  6. Iterate steps 2–5 until stable.
- This alternating scheme **isolates sparsity vs. smoothness** while ensuring accuracy + interpretability.

K-Fold CV algorithm for Sparsity

K-Fold CV (Row Sparsity)

1. Split $\boldsymbol{X} \in \mathbb{R}^{n \times M}$ into K column groups $\{\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(K)}\}$.

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    3. Fold error:

$$\mathrm{Err}_j^{(k)} = \frac{1}{\tilde{M}} \|\boldsymbol{X}^{(k)} - u_j^{(-k)} (v_j^{(k)})^\top\|_F^2.$$

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Variance Explained: Classical vs Regularized FPCA

- **Classical FPCA:** Loadings v_j orthonormal; scores $u_j = Xv_j$ uncorrelated. Variance explained by first J PCs:

$$\sum_{j=1}^{J} \|u_j\|^2 = \mathrm{trace}\big(V_J^\top X^\top X\, V_J\big), \qquad V_J = [v_1, \ldots, v_J].$$

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- **Issue under regularization:** smoothness/sparsity break orthogonality $\rightarrow$ scores become correlated $\rightarrow$ naive sum $\sum \|u_j\|^2$ **double-counts** variance (cf. Huang et al., 2008).

Subspace-Projection Definition of Explained Variance

- Normalize loadings and stack:

$$V_J = [v_1, \ldots, v_J], \qquad v_j \leftarrow \frac{v_j}{\|v_j\|}.$$

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- Projected data and explained variance:

$$X_J = XH_J, \qquad \mathrm{V}_{\mathrm{tot}} = \mathrm{tr}(X^\top X), \qquad \mathcal{V}_J = \|X_J\|_F^2 = \mathrm{tr}(H_J X^\top X H_J).$$

PVE, Incremental PVE, and Properties

- Incremental variance:

$$\Delta \mathcal{V}_j = \mathcal{V}_j - \mathcal{V}_{j-1}, \qquad \mathcal{V}_0 = 0.$$

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$$\mathrm{PVE}(J) = \frac{\mathcal{V}_J}{\mathsf{V}_{\mathsf{tot}}}, \qquad \mathrm{pve}_j = \frac{\Delta\mathcal{V}_j}{\mathsf{V}_{\mathsf{tot}}} = \mathrm{PVE}(j) - \mathrm{PVE}(j-1).$$

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- **Key properties:**
  - No double-counting (works with correlated scores).
  - Reduces to classical PCA when $V_J^\top V_J = I_J$.
  - Monotone in J ($\mathcal{V}_J$ increases).
  - $\Delta\mathcal{V}_j$ = unique variance added by component j.

Simulation: Two-way Functional Data

- **Data-generating process:**
  $X_{ij}^{(1)} = u_{i1} v_{11}(t_j) + u_{i2} v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1} v_{21}(t_j) + u_{i2} v_{22}(t_j) + \epsilon_{ij}^{(2)},$
- **Latent scores:** generated as smooth functions

$$u_1(s) = \begin{cases} \sin(\pi s), & s > 0, \\ 0, & \text{otherwise,} \end{cases} \qquad u_2(s) = \sin(2\pi s), \quad s \in [-1, 1].$$

- **Functional PCs:**
  - Variable 1: $v_{11}(t) = \frac{t+\sin(\pi t)}{s_1}, \qquad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$
  - Variable 2:

$$v_{21}(t) = \begin{cases} \dfrac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise,} \end{cases} \qquad v_{22}(t) = \begin{cases} \dfrac{\sin(2\pi t)}{s_4}, & t \le -\frac{1}{3}, \\ \dfrac{\sin(\pi t)}{s_4}, & t \ge \frac{1}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Evaluation Metrics

- **Integrated Squared Error (ISE):**
  For replicate r and component u_1:

$$\mathrm{ISE}_r^{(u_1,\text{method})} = \frac{1}{m}\sum_{j=1}^{m}\left(u_1(t_j) - \widehat{u}_1^{(\text{method})}(t_j)\right)^2.$$

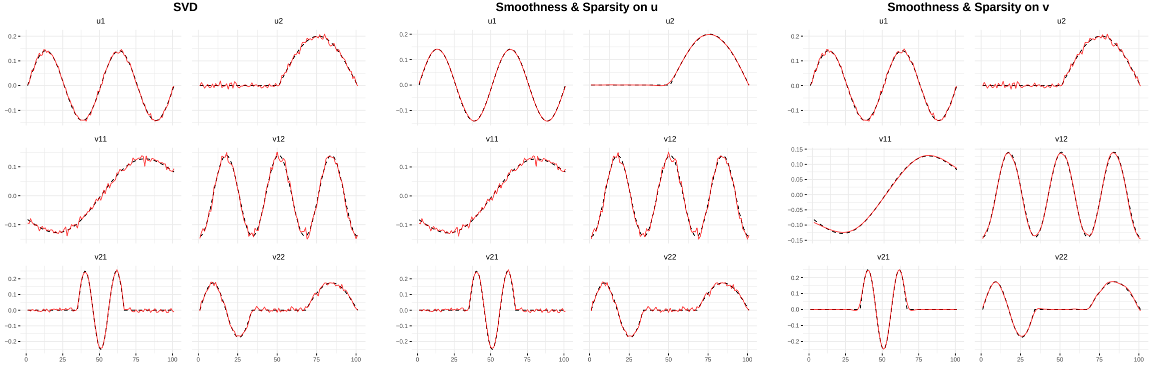
- **Relative ISE (R_ISE):** ratio vs best method

$$R_r^{(u_1,\text{method})} = \frac{\mathrm{ISE}_r^{(u_1,\text{method})}}{\mathrm{ISE}_r^{(u_1,\text{best})}}.$$

- **Monte Carlo averages:**

$$\overline{R}^{(u_1,\text{method})} = \frac{1}{N}\sum_{r=1}^{N} R_r^{(u_1,\text{method})}, \qquad \mathrm{SE}(\overline{R}) = \sqrt{\frac{1}{N(N-1)}\sum_{r=1}^{N}\left(R_r - \overline{R}\right)^2}.$$

Simulation Results



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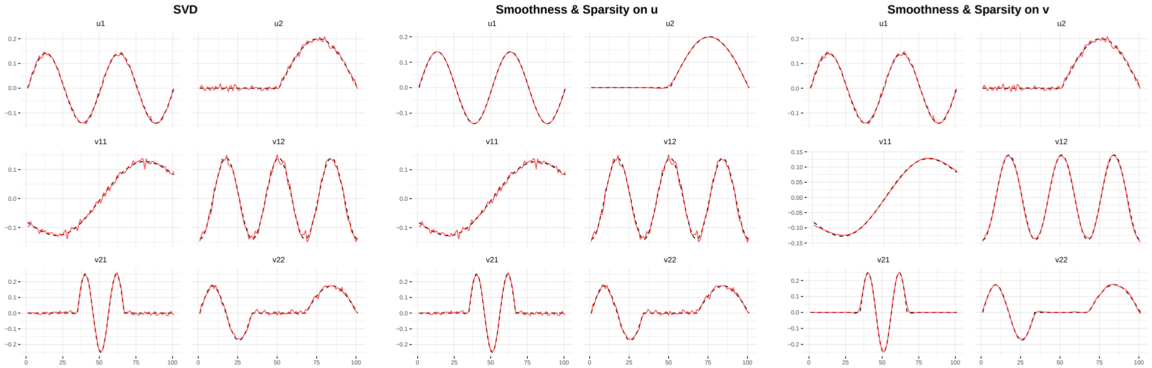


Simulation Results

Table 3: Mean ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	0.3651	0.1587	0.00005	0.00010
Smooth+Sparse *v*	0.3651	0.1587	**0.00002**	**0.00002**
Smooth+Sparse *u*	**0.3650**	**0.1584**	0.00005	0.00010
Two-way Smoothness	**0.3650**	0.1585	**0.00002**	**0.00002**
Two-way Sparsity	0.3651	0.1586	0.00004	0.00009
Two-way Sm+Sp	**0.3650**	**0.1584**	**0.00002**	**0.00002**

Table 4: Mean Relative ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	1.000	1.001	8.21	5.23
Two-way Sparsity	1.000	1.001	7.71	4.61
Smooth+Sparse *v*	1.000	1.001	1.05	**1.01**
Smooth+Sparse *u*	**1.000**	**1.000**	8.19	5.21
Two-way Smoothness	**1.000**	**1.000**	**1.00**	1.21

- Two-way Smooth+Sparse consistently yields the lowest errors across u and v.

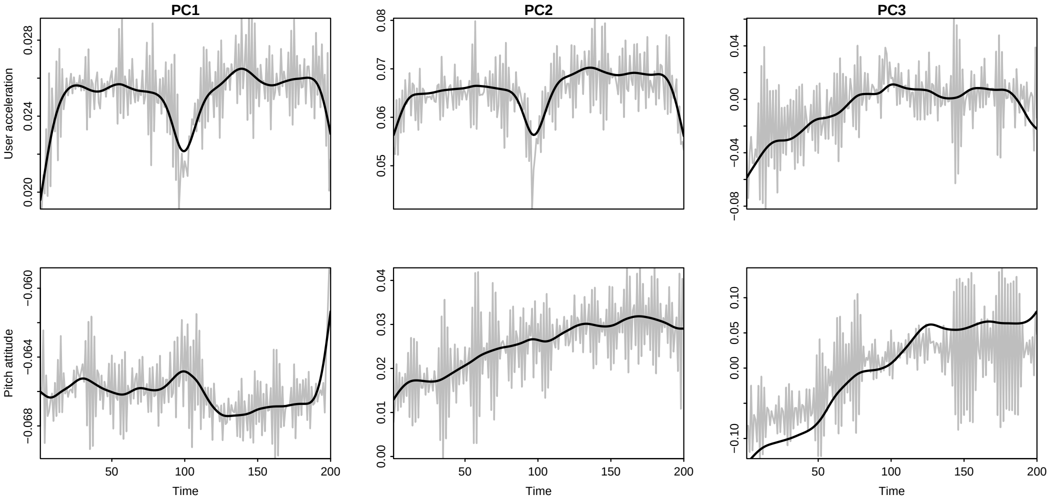
Application: Motion Sense Data

- **Dataset:** Acceleration and pitch from 24 people, 4 activities (jogging, walking, sitting, standing), about 2–3 min each.
- **Goal:** Compare SVD vs **two-way sparse + smooth ReMFPCA** on these multivariate functional signals.
- **Rescaling (Happ & Greven, 2018):** balance variables so each contributes equally.

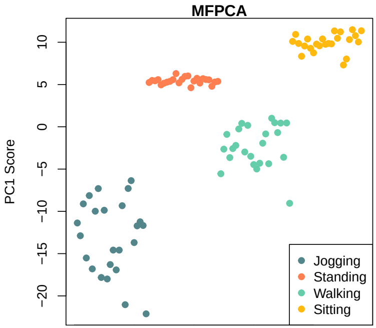
$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^{m} \widehat{\mathrm{Var}}(X_j(t_i)) \right)^{-1}, \qquad \tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i).$$

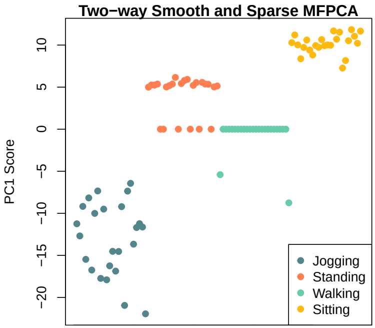
- **Penalties used:** smoothness + sparsity on loadings v; sparsity on scores u (treated as random effects).

Results: Functional PCs (SVD vs ReMFPCA)



Results: PC Scores and Interpretation





- Sparsity on scores: **PC1 scores for walking about 0** (partially standing too).
- Interpretation: walking contributes little to PC1; removing it **improves interpretability** without hurting fit.
- Takeaway: **Two-way smooth + sparse ReMFPCA** yields cleaner PCs and activity-informative scores.

Conclusion & Future Work

- **Unified FPCA Framework**
  - Combines **smoothness** (denoise, interpretability) + **sparsity** (variable selection).
  - Extends from **univariate** → **multivariate** → **two-way functional data**.
- **Methodology**
  - Penalized SVD with roughness + ℓ_1 penalties.
  - Two-way regularization: smoothness & sparsity on both **scores (*u*)** and **loadings (*v*)**.
  - Efficient parameter tuning: **conditional GCV** & **K-fold CV** (with 1-SE rule).
- **Results**
  - Simulations & applications (mortality, call-center, image data).
  - Outperforms one-way or single-penalty methods.
  - Produces **low-rank, denoised, interpretable components**.

Accessible Implementation: R Package & Future Work

- Implemented in **R package `ReMPCA` (GitHub)**
  - Univariate & multivariate FPCA with penalties.
  - Two-way MFPCA for matrix-valued functions.
  - Automated tuning (CV, GCV, 1-SE rule).
  - Diagnostic tools: variance explained, visualization, heatmaps.
  - Early support for **hybrid data (scalar + functional + image)**.
- **Hybrid Data Extensions**
  - Image–Functional Hybrid PCA $\rightarrow$ simultaneous dimension reduction.
  - Scalar–Functional Integration $\rightarrow$ joint low-dim space.
  - Nonlinear Extensions $\rightarrow$ kernel FPCA, neural nets.
- **Applications:**
  - Neuroimaging
  - Personalized medicine
  - Environmental monitoring

  Takeaway: Smooth + sparse + two-way FPCA offers a **theoretical foundation, practical algorithms, and open software** to enable next-generation functional data analysis.

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